

AQA A-Level Further Mathematics (7367)

Question Paper

Continuous Random Variables

SD3 – Find median + quartiles

(A-Level)

Name: _____

Class: _____

Date: _____

Time: **327 min.**

Marks: **261 marks**

Comments:

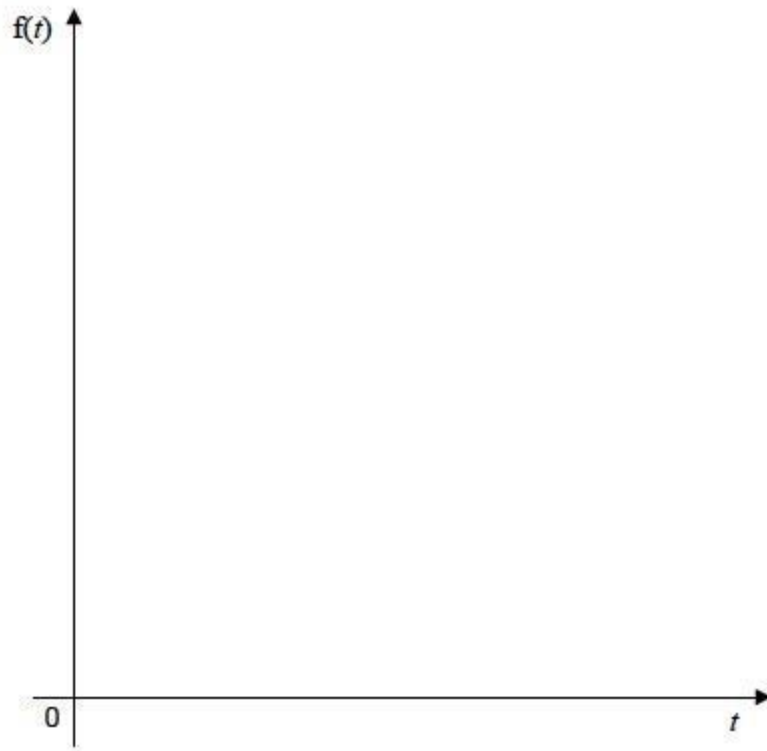
© 2024 Exams Papers Practice. All Rights Reserved

Q1.

The continuous random variable T has probability density function defined by

$$f(t) = \begin{cases} \frac{1}{3} & 0 \leq t \leq \frac{3}{2} \\ \frac{9-2t}{18} & \frac{3}{2} \leq t \leq \frac{9}{2} \\ 0 & \text{otherwise} \end{cases}$$

- (a) (i) Sketch this probability density function below.



(2)

- (ii) State the median of T .

(1)

- (b) (i) Find $E(T)$

(2)

- (ii) Given that $E(T^2) = \frac{15}{4}$, find $\text{Var}(4T - 5)$

(3)

(Total 8 marks)

Q2.

A continuous random variable X has probability density function defined by



$$f(x) = \begin{cases} kx^2 & 0 \leq x \leq 3 \\ 9k & 3 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) Show that the value of k is $\frac{1}{18}$.

(4)

- (c) (i) Write down the median value of X .

- (ii) Calculate the value of the lower quartile of X .

(4)

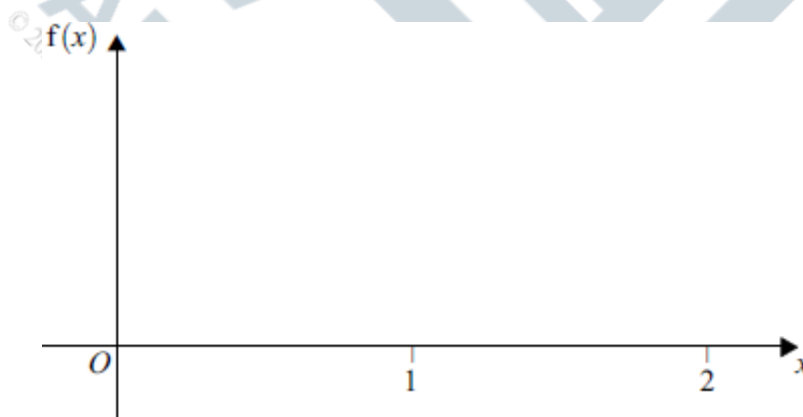
(Total 11 marks)

Q3.

A continuous random variable X has the probability density function defined by

$$f(x) = \begin{cases} x^2 & 0 \leq x \leq 1 \\ \frac{1}{3}(5 - 2x) & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f on the axes below.



(3)

- (b) (i) Find the cumulative distribution function, F , for $0 \leq x \leq 1$.

(2)

- (ii) Hence, or otherwise, find the value of the lower quartile of X .

(2)

- (c) (i) Show that the cumulative distribution function for $1 \leq x \leq 2$ is defined by

$$F(x) = \frac{1}{3} (5x - x^2 - 3)$$

(4)

- (ii) Hence, or otherwise, find the value of the upper quartile of X .

(4)

(Total 15 marks)

Q4.

The random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{40}(x+7) & 1 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(2)

- (b) Find the **exact** value of $E(X)$.

(3)

- (c) Prove that the distribution function F , for $1 \leq x \leq 5$, is defined by

$$F(x) = \frac{1}{80}(x+15)(x-1)$$

(4)

- (d) Hence, or otherwise:

- (i) find $P(2.5 \leq X \leq 4.5)$;

(2)

- (ii) show that the median, m , of X satisfies the equation $m^2 + 14m - 55 = 0$.

(3)

- (e) Calculate the value of the median of X , giving your answer to three decimal places.

(2)

(Total 16 marks)

Q5.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{3}{8}x^2 & 0 \leq x \leq \frac{1}{2} \\ \frac{3}{32} & \frac{1}{2} \leq x \leq 11 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) Show that:

$$(i) \quad P(X \geq 8) = \frac{1}{4}$$

$$(ii) \quad P(X \geq 3) = \frac{3}{4}$$

(3)

(c) Hence write down the **exact** value of:

(i) the interquartile range of X ;

(ii) the median, m , of X .

(3)

(d) Find the **exact** value of $P(X < m \mid X \geq 3)$.

(3)

(Total 12 marks)

Q6.

The continuous random variable X has the probability density function defined by

$$f(x) = \begin{cases} \frac{3}{8}(x^2 + 1) & 0 \leq x \leq 1 \\ \frac{1}{4}(5 - 2x) & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

(a) The cumulative distribution function of X is denoted by $F(x)$.

Show that, for $0 \leq x \leq 1$,

$$F(x) = \frac{1}{8}x(x^2 + 3)$$

(3)

(b) Hence, or otherwise, verify that the median value of X is 1.

(2)

(c) Show that the upper quartile, q , satisfies the equation $q^2 - 5q + 5 = 0$ and hence that $q = \frac{1}{2}(5 + \sqrt{5})$.

(5)

(d) Calculate the **exact** value of $P(q < X < 1.5)$.

(4)

Q7.

The random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 1 \\ \frac{1}{8}(x-4)^2 & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) State values for the median and the lower quartile of X .

(2)

- (b) Show that, for $1 \leq x \leq 4$, the cumulative distribution function, $F(x)$, of X is given by

$$F(x) = 1 + \frac{1}{54}(x-4)^3$$

(You may assume that $\int (x-4)^2 dx = \frac{1}{3}(x-4)^3 + c$.)

(4)

- (c) Determine $P(2 \leq X \leq 3)$.

(2)

- (d) (i) Show that q , the upper quartile of X , satisfies the equation $(q-4)^3 = -13.5$.

(3)

- (ii) Hence evaluate q to three decimal places.

(1)

(Total 12 marks)

Q8.

The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 1 \\ \frac{3-x}{4} & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) Explain why the value of η , the median of X , is 1.

(2)

- (c) Show that the value of μ , the mean of X , is $\frac{13}{12}$.

(4)

- (d) Find $P(X < 3\mu - \eta)$.

(3)

(Total 12 marks)

Q9.

The continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{x+1}{k+1} & -1 \leq x \leq k \\ 1 & x > k \end{cases}$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for $P(X < 0)$.

(2)

- (b) Determine an expression, in terms of k , for the lower quartile, q_1 .

(3)

- (c) Show that the probability density function of X is defined by

$$f(x) = \begin{cases} \frac{1}{k+1} & -1 \leq x \leq k \\ 0 & \text{otherwise} \end{cases}$$

(2)

- (d) Given that $k = 11$:

- (i) sketch the graph of f ;

(2)

- (ii) determine $E(X)$ and $\text{Var}(X)$;

(2)

- (iii) show that $P(q_1 < X < E(X)) = 0.25$.

(2)

(Total 13 marks)

Q10.

Engineering work on the railway network causes an increase in the journey time of commuters travelling into work each morning.

The increase in journey time, T hours, is modelled by a continuous random variable with probability density function

$$f(t) = \begin{cases} 4t(1-t^2) & 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that $E(T) = \frac{8}{15}$. (3)
- (b) (i) Find the cumulative distribution function, $F(t)$, for $0 \leq t \leq 1$. (2)
- (ii) Hence, or otherwise, for a commuter selected at random, find $P(\text{mean} < T < \text{median})$ (5)
- (Total 10 marks)

Q11.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{5}(2x+1) & 0 \leq x \leq 1 \\ \frac{1}{15}(4-x)^2 & 1 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (2)
- (b) (i) Show that the cumulative distribution function, $F(x)$, for $0 \leq x \leq 1$ is $F(x) = \frac{1}{5}x(x+1)$ (3)
- (ii) Hence find $P(X \geq 0.5)$. (2)
- (iii) Verify that the lower quartile, q_1 , satisfies $0.5 < q_1 < 0.75$ (3)
- (Total 10 marks)

Q12.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{5}(2x+1) & 0 \leq x \leq 1 \\ \frac{1}{15}(4-x)^2 & 1 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(2)

- (b) (i) Show that the cumulative distribution function, $F(x)$, for $0 \leq x \leq 1$ is

$$F(x) = \frac{1}{5}x(x+1)$$

(3)

- (ii) Hence write down the value of $P(X \leq 1)$.

(1)

- (iii) Find the value of x for which $P(X \geq x) = \frac{17}{20}$.

(5)

- (iv) Find the lower quartile of the distribution.

(4)

(Total 15 marks)

Q13.

The time, T hours, that the supporters of Bracken Football Club have to queue in order to obtain their Cup Final tickets has the following probability density function.

$$f(t) = \begin{cases} \frac{1}{5} & 0 \leq t < 3 \\ \frac{1}{45}t(6-t) & 3 \leq t \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) Write down the value of $P(T = 3)$.

(1)

- (c) Find the probability that a randomly selected supporter has to queue for at least 3 hours in order to obtain tickets.

(2)

- (d) Show that the median queuing time is 2.5 hours.

(2)

- (e) Calculate $P(\text{median} < T < \text{mean})$.

(6)

(Total 14 marks)

Q14.

The error, X millimetres, made when the heights of prospective members of a new gym club are measured can be modelled by a rectangular distribution with the following probability density function.

$$f(x) = \begin{cases} k & -4 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

- (a) State the value of k . (1)
 - (b) Write down the value of $E(X)$. (1)
 - (c) Calculate $P(X > 0)$. (2)
 - (d) The height of a randomly selected prospective member is measured. Find the probability that the **magnitude** of the error made exceeds 3.5 millimetres. (3)
- (Total 7 marks)**

Q15.

The discrete random variable X has

$$E(X) = 4 \quad \text{and} \quad E(X^2) = 17.2.$$

- (a) Find the value of $\text{Var}(X)$. (2)
 - (b) A circle has radius $(X + 8)$.

Find, in terms of π , values for the mean and variance of the **circumference**, C , of the circle. (4)
 - (c) The **area**, S , of the circle, with radius $(X + 8)$, is to be expressed in the form
$$\pi(X^2 + aX + b),$$

where a and b are positive constants.
 - (i) Find values for a and b . (2)
 - (ii) Hence find, in terms of π , the value of $E(S)$. (2)
- (Total 10 marks)**

Q16.

At a Post Office, the time, T minutes, that customers have to wait in order to be served has the following probability density function.

$$f(t) = \begin{cases} \frac{t^2}{18} & 0 \leq t < 3 \\ \frac{1}{4}(5-t) & 3 \leq t \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Write down the value of $P(T = 4)$. (1)
- (b) Show that the median waiting time is 3 minutes. (2)
- (c) Find the probability that customers have to wait for less than 3.5 minutes to be served. (5)
- (d) Calculate the mean time that customers have to wait in order to be served. (5)
- (Total 13 marks)**

Q17.

All consultations with an optician are by appointment. The time, T minutes by which an appointment is delayed has the following probability density function.

$$f(t) = \begin{cases} \frac{1}{54}t^2 & 0 \leq t < 3 \\ \frac{1}{6} & 3 \leq t < 6 \\ \frac{1}{24}(10-t) & 6 \leq t < 10 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (4)
- (b) Use your graph to find:
- (i) $P(3 \leq T < 6)$; (2)
- (ii) $P(T > 6)$; (2)
- (iii) the lower quartile (4)
- (c) What proportion of appointments is delayed by less than 2 minutes? (4)
- (Total 16 marks)**

Q18.

The delay, T minutes, in connecting electronically to a ticket booking service can be modelled by the distribution function:

$$P(T \leq t) = 1 - e^{-2t}, \quad t \geq 0.$$

- (a) (i) Write down the probability density function of T . (1)
- (ii) State values for the mean and standard deviation of T . (2)
- (b) Calculate the probability that the delay exceeds 3 minutes. (2)
- (c) Find the median delay, giving your answer in seconds, to the nearest second. (3)
- (Total 8 marks)**

Q19.

The total amount of added time, S minutes, played per match by a team in a local football league is distributed with probability density function given by:

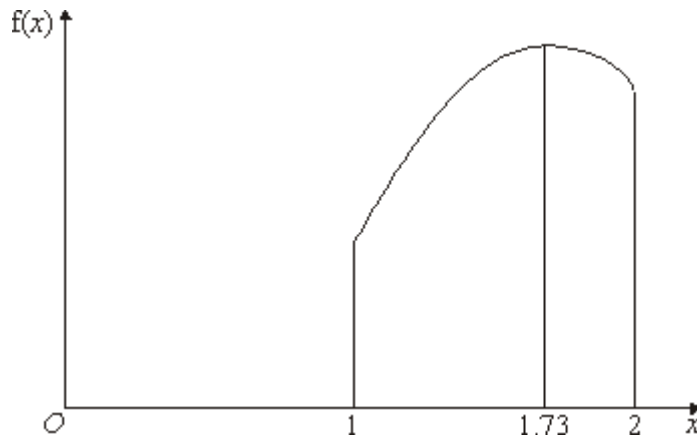
$$f(s) = \begin{cases} ks^2 & 1 \leq s \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that the exact value of the constant k is $\frac{3}{124}$. (4)
- (b) Determine $P(S < 3)$. (3)
- (c) Calculate the median amount of added time played per match, giving your answer to the nearest minute. (4)
- (Total 11 marks)**

Q20.

The sketch below shows the probability density function $f(x)$ of the continuous random variable X .

($f(x) = 0$ outside the interval $1 \leq x \leq 2$.)



You are given that $\int_1^{1.58} f(x) dx = 0.5$ $\int_{1.73}^2 f(x) dx = 0.291$

$$\int_1^2 xf(x) dx = 1.54 \quad \int_1^2 (x - 1.54)^2 f(x) dx = 0.0751$$

(a) Write down the value of:

(i) $\int_1^2 f(x) dx$;

(ii) the mean of X ;

(iii) the median of X ;

(iv) the mode of X ;

(v) the range of X .

(5)

(b) Find:

(i) the probability that X lies between the median and the mode;

(2)

(ii) the standard deviation of X ;

(2)

(iii) $E(X^2)$.

(2)

(Total 11 marks)

Q21.

The random variable, X , has probability density function

$$f(x) = \begin{cases} cx + d & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases} \quad \text{where } c \text{ and } d \text{ are constants}$$

- (a) Show that $2c + 2d = 1$. (3)
- (b) Find the median of X :
- (i) when $c = 0$ and $d = 0.5$; (2)
- (ii) when $c = 0.5$ and $d = 0$. (3)
- (c) The following pairs of values are suggested for c and d .

Suggestion A: $c = 0.5$ and $d = 0.5$

Suggestion B: $c = -0.5$ and $d = 1$

Suggestion C: $c = 1.5$ and $d = -1$

Only one of the suggestions provides a valid probability density function.

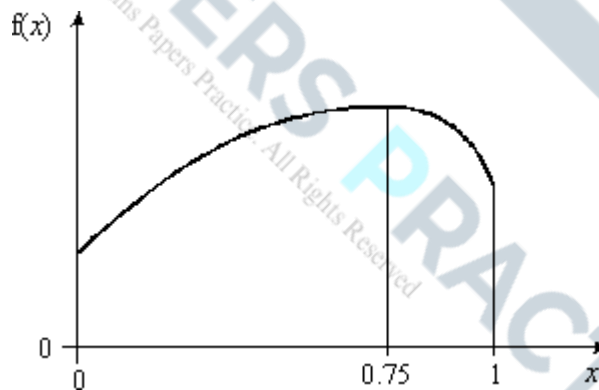
For **each** suggestion state whether or not it is valid. If you state it is not valid, give a reason.

(5)
(Total 13 marks)

Q22.

The sketch below shows the probability density function $f(x)$ of the continuous random variable X .

$[f(x) = 0 \text{ outside the interval } 0 \leq x \leq 1]$



- (a) Explain why the value of the mode of X is 0.75. (1)
- (b) You are given that:

$$E(X) = 0.542$$

$$E(X^2) = 0.369$$

$$\int_0^{0.557} f(x) dx = 0.500$$

$$\int_0^{0.542} f(x) dx = 0.483$$

$$\int_0^{0.75} f(x) dx = 0.719$$

- (i) Find the standard deviation of X . (3)
 - (ii) Find the probability that X exceeds the mode. (2)
 - (iii) Write down the value of the median of X . (1)
 - (iv) Evaluate the probability that X lies between the median and the mean. (3)
- (Total 10 marks)

